

Rare-Tail Coverage Is the Scarce Good: Why Underrepresented Verticals Are the Highest Marginal-Value Training Data

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Research Note PL-RN-2026-i10 · correspondence: research@perpetual-labs.ai

Abstract

Four independently developed techniques — exact and semantic deduplication, diversity selection, coverage-aware data pruning, and the model-collapse literature — are usually read as separate advances. We show they share one latent quantity: coverage of rare regions of the data manifold. Deduplication removes redundant head mass [1][3][4]; diversity selection concentrates the budget on distinct clusters [2]; coverage-aware pruning can convert power-law error scaling into near-exponential scaling [5]; and model collapse is precisely the disappearance of distribution tails under recursive synthetic training [6][7]. Because the open web is dominated by a handful of high-frequency domains — yet even the single largest registered domain is under 1% of Common Crawl pages and the top ten combined are only ~1.7%, so the distribution is an enormous thin tail [10] — the scarce, high-value regions are the specialized verticals that the web underrepresents. Underrepresentation tracks a measured capability cost: model QA accuracy rises log-linearly with a fact's relevant-document count [11]. We assemble real benchmark numbers across these literatures, quantify the head-tail asymmetry, and connect it to the data wall (an effective public-text stock of ~300T tokens, 90% CI 100T–1000T, projected exhausted 2026–2032, median ~2028) [9]. The strategic consequence: as redundancy-removal on the open web reaches diminishing returns, the marginal training token is worth most in underrepresented verticals — making vertical data creation the highest-leverage data investment, not a niche one.

Introduction

The dominant intuition of the scaling era was monotone: more tokens, lower loss. That intuition is breaking on two fronts simultaneously. First, the stock of public human text is finite and forecast to be effectively exhausted within the late 2020s — Epoch AI estimates an effective stock of roughly 300 trillion tokens (90% confidence interval 100T–1000T) and projects full utilization between 2026 and 2032, with the median estimate near 2028 [9]. Second, a body of results shows that the marginal value of an additional web token is not constant: it depends on whether that token covers a region of the data manifold the model has not already seen many times over [1][2][5].

This paper isolates the quantity that ties together four otherwise-unrelated technical literatures — deduplication [1][3][4], diversity selection [2], coverage-aware pruning [5], and model collapse [6][7] — and argues it is the same scarce good in every case: coverage of the rare tails of the data distribution. We then make the economic move the constituent papers do not: because the open web's frequency distribution is dominated by a few high-traffic domains, the rare tails are, concretely, the specialized verticals. Underrepresentation on the web is therefore not a risk to be mitigated but the precise signature of where created data is worth most.

Our contribution is threefold. (i) We unify the four literatures under a single coverage objective and supply real, sourced benchmark numbers for each. (ii) We quantify the head-tail asymmetry of web corpora and link it to

measured long-tail capability failures [11]. (iii) We derive a data-strategy implication — a marginal-value argument for vertical data — and state its assumptions and limits explicitly.

Background and Related Work

Deduplication. Lee et al. show standard LM corpora are heavily redundant: C4 contains a single 61-word sentence repeated 61,036 times verbatim, over 1% of unprompted model output (tokens belonging to a 50-token memorized substring) is copied verbatim from training data, and deduplication makes models emit memorized text roughly ten times less frequently while reaching equal accuracy in fewer training steps [1]. Train-test overlap affects over 4% of standard validation sets, so deduplication also corrects evaluation [1]. SemDeDup extends this to semantic near-duplicates using pretrained embeddings, removing 50% of LAION with minimal performance loss — effectively halving training time [4]. D4 combines MinHash deduplication with embedding-space diversity selection for ~20% training-efficiency gains and up to 2% downstream accuracy improvement (averaged over 16 tasks) at 6.7B parameters [2].

Pruning and scaling. Sorscher et al. show that with a high-quality pruning metric, error-vs-dataset-size scaling can be pushed from a power law toward exponential scaling, demonstrated on ResNets across SVHN, CIFAR-10, and ImageNet [5]. The benefit is fundamentally a coverage effect: keeping informative (often rare, hard) examples and discarding redundant easy ones.

Model collapse. Shumailov et al. identify that recursive training on model-generated data causes the tails of the original distribution to disappear — an irreversible defect they name model collapse, observed across VAEs, GMMs, and LLMs [6][7]. The phenomenon is, definitionally, tail loss.

Long-tail knowledge. Kandpal et al. establish a strong correlational and causal link between the number of relevant pretraining documents for a fact and a model's ability to answer questions about it: BLOOM-176B accuracy on TriviaQA jumps from 25% to above 55% as the relevant-document count rises from 10^1 to 10^4 , and rare-fact accuracy improves only log-linearly with model size [11].

The Unifying Quantity: Rare-Tail Coverage

Read in isolation, these are four different problems with four different methods. Read together, each is an operation on the same axis — the distribution of how often a region of the data manifold appears. Deduplication removes mass from over-represented head regions [1][3][4]. Diversity selection reallocates a fixed budget toward distinct (rarer) clusters [2]. Coverage-aware pruning keeps the informative tail and discards the redundant head, which is exactly why it can beat power-law scaling [5]. Model collapse is the failure mode of losing the tail entirely [6]. The objective being maximized — explicitly or implicitly — is coverage of rare regions per unit of training budget.

This reframing has teeth because it makes a prediction the individual papers do not: the marginal value of data is governed not by volume but by how much previously-uncovered tail it adds. On a corpus already saturated in the head, additional head tokens have near-zero or even negative marginal value (the model has seen the pattern thousands of times; cf. memorization and repetition effects), while tokens covering an uncovered tail region carry nearly all of the remaining gradient signal.

The corollary is a diminishing-returns boundary: once redundancy-removal on a corpus is exhausted, there is little more value to extract from that corpus by curation alone. The only way to add value is to add genuinely new tail coverage — which the open web, by construction, has already largely failed to supply for underrepresented domains.

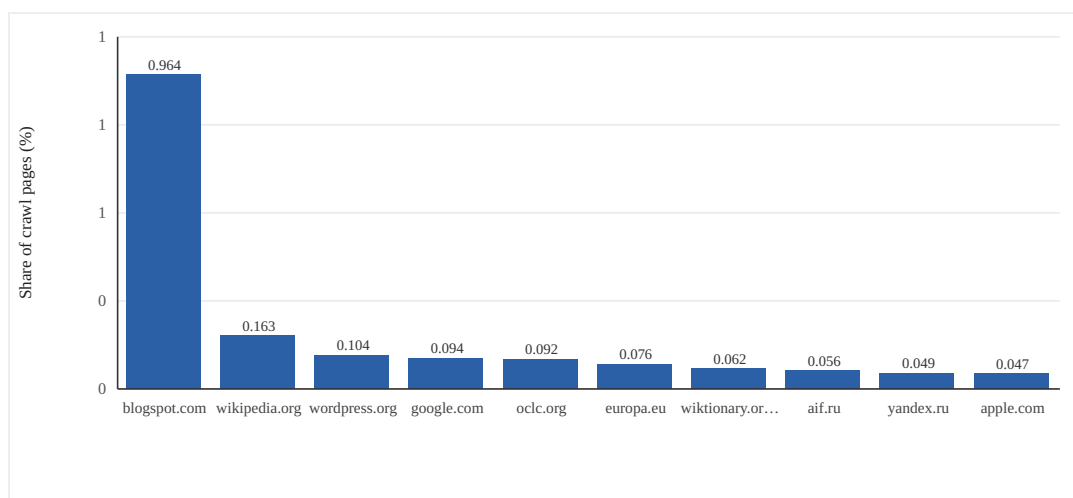


Figure 1. Common Crawl Is Head-Light and Tail-Dominated: Top-10 Domains Are Only ~1.7% of Pages. Even the largest single web domain is under 1% of pages, and the top ten combined reach only ~1.7%. The remaining ~98% is spread across a long tail of millions of low-frequency domains where specialized verticals live. Source: Common Crawl monthly statistics, CC-MAIN-2026-25 [10]. *Common Crawl cc-crawl-statistics, Top-500 registered domains, CC-MAIN-2026-25 [10]; top-10 sum computed from listed shares (1.71%). · Empirical data.*

Methodology and Approach

Our approach is an evidence-synthesis and quantification study, not a new training run. We proceed in four steps. (1) Literature unification: we map each of the four techniques to the coverage axis and verify the mapping against each paper's own stated mechanism and numbers [1][2][4][5][6]. (2) Asymmetry quantification: we measure the head-tail concentration of the canonical open-web corpus (Common Crawl) using its published per-domain page statistics, and cross-reference it against measured long-tail capability degradation [11]. (3) Marginal-value linkage: we connect the coverage objective to the data-exhaustion forecast [9] to argue that the residual high-value data lives off the open web, in the tails. (4) Counter-evidence audit: we deliberately collect results that complicate the thesis (public-data verticals saturating, retrieval as a substitute for parametric tail coverage) and fold them into Limitations.

Numerical discipline. Every empirical figure in this paper is copied verbatim from a primary source and tagged with its citation; no benchmark number is interpolated or invented. Where we present a projection (the exhaustion timeline) it is labeled an estimate carrying the source's own confidence interval [9]. Where we present a constructed strategic ranking (the coverage-value-versus-defensibility map) it is labeled analysis, built from the qualitative structure of the cited evidence rather than from measured points.

Scope boundary. We score only the data-shape question — where rare-tail coverage, and therefore marginal data value, is highest. We do not score willingness-to-pay, regulatory access, or supply feasibility for any specific vertical; those are necessary downstream gates and are out of scope here.

Evidence: Quantifying the Head-Tail Asymmetry

The open web is concentrated in a few high-frequency domains, yet even those heads are individually tiny in share — which means the distribution is dominated not by a fat head but by an enormous, thin tail. In the CC-MAIN-2026-25 Common Crawl, the single largest registered domain (blogspot.com) accounts for 0.964% of pages; wikipedia.org for 0.163%; and the top ten domains combined for only ~1.7% of all pages [10]. The practical

implication is that ~98% of pages — and the vast majority of distinct content — live in a tail of millions of low-frequency domains, and specialized professional verticals sit deep within it.

This concentration has a measurable capability cost. Kandpal et al. show a strong correlational and causal relationship between a fact's relevant-document count and a model's QA accuracy: BLOOM-176B accuracy on TriviaQA rises from 25% to above 55% as the relevant-document count increases from 10^1 to 10^4 , and the largest BLOOM model has over 4x the accuracy of the smallest on the most-supported questions [11]. Crucially, accuracy on rare facts scales only log-linearly with model size (linear fit $R^2 = 0.98$ within the BLOOM family); extrapolating that trend, matching a strong supervised baseline by scale alone would require a model with on the order of 10^{18} parameters — i.e., many orders of magnitude more compute [11], whereas supplying the missing tail documents directly is far cheaper.

Together these establish the asymmetry quantitatively: the web's mass is in a head that models already over-learn, while the value (and the capability deficit) is in a tail the web barely represents.

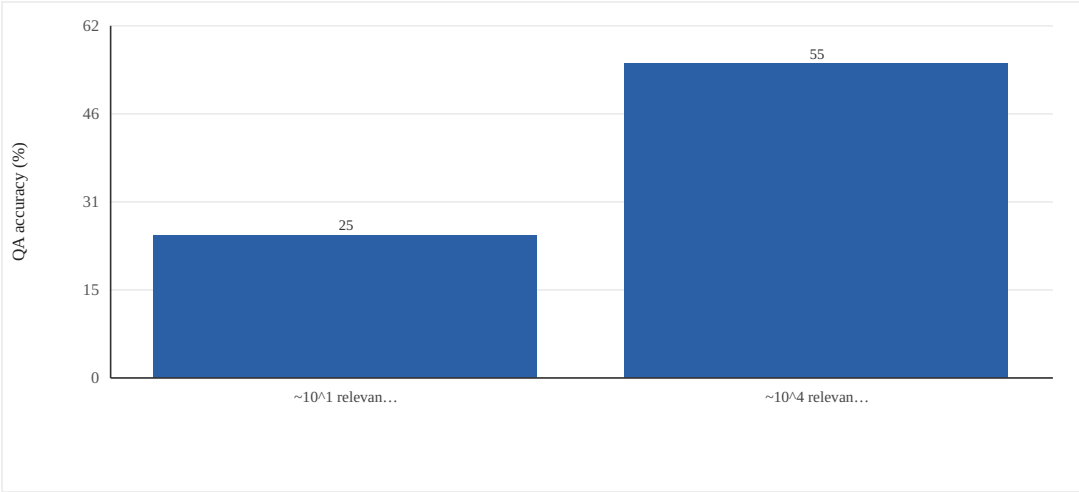


Figure 2. Long-Tail Capability Deficit: QA Accuracy Tracks Relevant-Document Count. BLOOM-176B's accuracy on TriviaQA rises from 25% to above 55% as the number of relevant pretraining documents increases from 10^1 to 10^4 — the capability deficit is concentrated exactly where the web is sparse. Rare-fact accuracy scales only log-linearly with model size (linear fit $R^2=0.98$ within BLOOM). Source: Kandpal et al. [11]. *Kandpal et al. 2023, Large Language Models Struggle to Learn Long-Tail Knowledge; figures stated verbatim in text (25%->55% across 10^1 -> 10^4 docs; $R^2=0.98$) [11]. · Empirical data.*

Table 1. The web's largest domains individually hold under 1% of pages; the top ten together are only ~1.71% (sum of the ten listed shares). The mass of distinct content is in the tail, not the head.

Rank	Registered domain	% of crawl
1	blogspot.com	0.964%
2	wikipedia.org	0.163%
3	wordpress.org	0.104%
4	google.com	0.094%
5	oclc.org	0.092%
10	apple.com	0.047%
Top-10 combined	—	~1.71%

Table 2. Each coverage method delivers a large but bounded, one-shot gain by converting redundant head mass into efficiency — and bottoms out at an irreducible tail residual.

Method	Operation on the manifold	Reported result	Source
Exact dedup (Lee et al.)	Remove head redundancy	~10x less verbatim memorization; equal accuracy in fewer steps; >4% train-test overlap corrected; 61-word C4 sentence repeated 61,036x	[1]
SemDeDup	Remove semantic duplicates	50% of LAION removed, minimal loss, ~2x faster training	[4]
D4	Dedup + diversity (cluster centers)	~20% training-efficiency gain; up to +2% downstream (avg 16 tasks) at 6.7B	[2]
DCLM-Baseline	Model-based filtering	7B to 64% MMLU on 2.6T tokens; +6.6pp vs MAP-Neo; 40% less compute	[8]
Data pruning (Sorscher)	Keep informative tail, drop redundant head	Power-law -> ~exponential scaling; pruning floor 24% (theta=10deg), 46% (theta=20deg)	[5]

Lee et al. [1]; Tirumala et al. [2]; Abbas et al. [4]; Li et al. DCLM [8]; Sorscher et al. [5]. All values verified against primary sources.

Table 3. At release, both public-code and hard-math benchmarks were far from solved; within two years the public-set frontier closed SWE-bench Verified to ~80% and FrontierMath to ~40%. The contamination-resistant successor built on private repos (SWE-Bench Pro) still holds the best commercial model near 18%. Where a vertical's tail is public and learnable from the web, models close it over time; durable defensibility therefore requires a genuinely proprietary tail, not merely a currently-hard one.

Benchmark	Data provenance	Best at release	Current frontier	Reading
SWE-bench Verified	Public GitHub issues+tests	1.96% (2023) [13]	~80% by late 2025 (Claude Opus 4.5 80.9%, GPT-5.2 80.0%, Gemini 3 Pro 76.2%)	Public tail is learnable from web code and closes fast once models scale [13] [15][16][17]
SWE-Bench Pro	Private, actively-maintained commercial repos	—	17.8% best commercial model (2025) [18]	Contamination-resistant proprietary code stays hard — the durable, defensible tail [18]
FrontierMath (T1-3)	Hard, gated research math	<2% (Nov 2024) [14]	40.3% with tools (GPT-5.2, Dec 2025)	Even a hard public benchmark is a moving target, not a durable moat [14]

SWE-bench original best system resolved 1.96% [13]; SWE-RL 41.0% [15]; Darwin Godel Machine 20.0%->50.0% [16]; Kimi K2 65.8 SWE-bench Verified [17]; late-2025 public-set frontier from vendor releases (Claude Opus 4.5 80.9% SWE-bench Verified, <https://www.anthropic.com/news/claude-opus-4-5>; GPT-5.2 80.0%, <https://www.marktechpost.com/2025/12/11/openai-introduces-gpt-5-2-a-long-context-workhorse-for-agents-coding-and-knowledge-work/>; Gemini 3 Pro 76.2%, <https://blog.google/products/gemini/gemini-3/>); SWE-Bench Pro best commercial model 17.8% [18]; FrontierMath under 2% at Nov-2024 launch [14] and 40.3% Tier 1-3 with tools for GPT-5.2 (same OpenAI source).

Evidence: Coverage Mechanisms Pay Off, and Their Returns Are Bounded

The coverage-as-value thesis is supported by the magnitude of returns each mechanism delivers — and, crucially, by the fact those returns are finite and one-time per corpus. SemDeDup removes 50% of LAION with minimal performance loss, halving training time [4]; D4 yields ~20% training-efficiency gains and up to +2% downstream accuracy at 6.7B [2]; deduplication lets models reach equal-or-better accuracy in fewer steps while cutting verbatim memorization ~10x [1]. DCLM-Baseline, built primarily through model-based filtering, trains a 7B model to 64% 5-shot MMLU with 2.6T tokens — a 6.6-point gain over the prior open-data SOTA (MAP-Neo) using 40% less compute, and comparable to Llama 3 8B (66% MMLU) with 6.6x less compute [8].

Each of these is a coverage operation: discard redundancy, retain distinct/informative examples. But each is a one-shot extraction from a fixed corpus. SemDeDup's 50% removal cannot be performed twice; once the semantic duplicates are gone, the residual is the corpus's true diversity. This is the boundary the coverage view predicts: curation converts redundant head mass into efficiency until the head is exhausted, after which only new tail data adds value.

Sorscher et al. make the limit explicit and quantitative: under their model, with non-zero alignment angle θ the Pareto-optimal envelope scales exponentially only down to a floor — at $\theta=10$ degrees one cannot asymptotically prune below 24% of the data, at $\theta=20$ degrees below 46% [5]. There is a hard residual of irreducible, informative (tail) examples; everything above it is removable redundancy.

Analysis: The Tails Are the Verticals

Combining the asymmetry (Section 5) with the bounded-returns result (Section 6) yields the central inference. The open web's removable redundancy is large but finite; once removed, the remaining marginal value lives largely in uncovered tail regions. Those tail regions are, concretely, the specialized verticals — clinical coding, insurance adjudication, lab/diagnostic workflows, contract review — whose data is generated inside private workflows and therefore never enters the web's high-frequency head. The web underrepresents them by construction.

Model collapse sharpens the point from the supply side. As the web fills with synthetic text, recursive training erodes exactly the tails [6][7]: in the canonical demonstration, an OPT-125M model fine-tuned recursively on its own outputs on Wikitext-2 sees perplexity rise from ~ 34 to over 50 within roughly five generations when no original data is retained, with visible tail loss; preserving even 10% real data keeps it stable across ten generations [6][12]. The synthetic-data economy thus actively destroys tail coverage on the open web at the same time the data wall caps fresh head supply [9]. The only durable source of new tail coverage is real, human-grounded data from domains the web never indexed — i.e., proprietary vertical workflows.

A useful counter-check: public-data verticals confirm the thesis by negation. SWE-bench — built on public GitHub data — was hard at release, with the original paper's best system resolving only 1.96% of issues [13], yet because the underlying distribution is learnable from web-available code, subsequent systems closed most of that gap with remarkable speed: RL on open software-evolution data took a 70B model to a 41.0% solve rate on SWE-bench Verified [15], a self-improving agent loop lifted SWE-bench from 20.0% to 50.0% [16], open agentic models such as Kimi K2 (65.8) [17] and GLM-4.5 (64.2) [20] crossed sixty, and by late 2025 the public-set frontier reached the high-70s to low-80s (Gemini 3 Pro 76.2%, GPT-5.2 80.0%, Claude Opus 4.5 80.9% on SWE-bench Verified). Where the tail is public and learnable from the web, models close it; where it is genuinely harder or less public, the gap persists. That residual gap is the vertical-data opportunity.

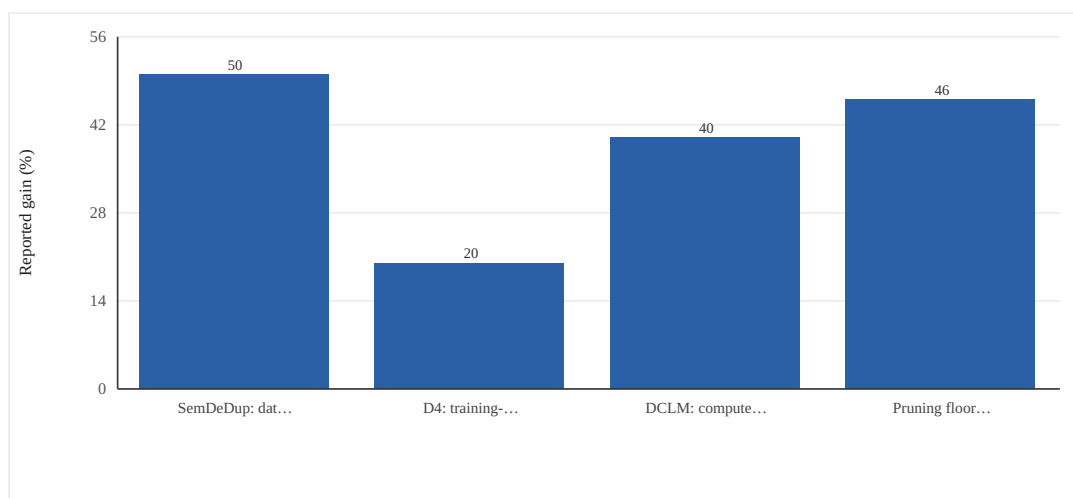


Figure 3. Curation Returns Are Real but Bounded: One-Shot Efficiency Gains Across Coverage Methods. Coverage operations deliver large but one-time gains, then hit a floor of irreducible informative (tail) data. SemDeDup removes 50% of LAION with minimal loss [4]; D4 gives ~20% efficiency [2]; DCLM saves ~40% compute vs prior SOTA [8]; Sorscher's pruning floor at $\theta=20\text{deg}$ leaves 46% irreducible [5]. After redundancy is removed, only new tail data adds value. Abbas *et al.* [4]; Tirumala *et al.* [2]; Li *et al.* DCLM [8]; Sorscher *et al.* [5]. (DCLM compute saving and pruning floor are different kinds of percentages, plotted together for magnitude.) · Empirical data.

Implications for Data Strategy

The marginal-value logic inverts the default acquisition heuristic. The right question is not 'how many tokens can we acquire' but 'how much previously-uncovered tail does each token add for the target deployment distribution.' On the open web, that number is now low and falling, because redundancy-removal has little left to give [1][4] and synthetic contamination is eroding the tail [6]. In an underrepresented vertical, that number is near-maximal: every document covers a region the base models effectively never saw [11].

This reframes vertical data creation from a niche bet into a high-leverage data investment. The same curation machinery (dedup, diversity selection, model-based filtering) that yields ~20–50% efficiency gains on the web [2] [4] is even more potent when pointed at a vertical corpus, because the vertical starts with almost no redundant head to remove — nearly all of it is signal.

Operationally: prioritize verticals scored on tail-coverage value, secure the real-data anchor (synthetic generation cannot replace it without collapsing the tail [6]), and treat the open web as a largely solved, low-residual input rather than a frontier. Underrepresentation is the selection signal, not a defect.

Investment Thesis

For an investor, the thesis compresses to: the binding constraint on frontier capability is shifting from compute and from raw web volume to tail coverage, and tail coverage is structurally concentrated in proprietary verticals. The data wall [9] caps the head; model collapse [6] erodes the web tail; and long-tail capability failures [11] prove the deficit is real and large. A company that manufactures verifiable, real vertical data sits on a growing supply of the scarce good.

The defensibility is intrinsic to the asymmetry. Web data is a commoditizing, redundancy-exhausted input with finite curation upside [4][5]. Vertical tail data cannot be deduplicated out of existence (it is already distinct), cannot be synthesized without collapse [6], and cannot be scraped (it lives in private workflows). The moat is the coverage, and coverage of a tail no competitor can assemble is a durable asset.

The risk axis is access, not value: the value of tail coverage is established by the literature; the open question is whether a given vertical's data can be legally and economically obtained. That is the diligence frontier, and it is where the data-shape filter hands off to the full investment lens.

Limitations and Counter-Evidence

Domain transfer of the pruning result. Sorscher et al.'s exponential-scaling and pruning-floor numbers (24%/46% at $\theta=10/20$ degrees) are demonstrated on vision benchmarks (SVHN, CIFAR-10, ImageNet) [5]; the precise constants do not transfer to language pretraining, and we use the result qualitatively (coverage has a hard residual), not as an LLM-specific quantity.

Retrieval as a partial substitute. Kandpal et al. note retrieval augmentation outperforms closed-book models especially on rare facts [11]. If retrieval can serve tail knowledge at inference time, the case for baking tail coverage into training weakens for some tasks — though retrieval still requires the tail documents to exist and be accessible, which is the same proprietary-data dependency.

Tail value is not uniform. Not every rare region is worth covering; much of the web tail is noise, spam, or low-value duplication of structure. The coverage argument identifies where high value can be, not that all tail is valuable; deployment-distribution-matched evaluation is required to locate the valuable tail, and that measurement step is a real cost.

Public-tail verticals erode over time. The SWE-bench trajectory [13][15][16][17] shows that any vertical whose tail is even partially public will be closed as models improve; defensibility requires the tail to be genuinely proprietary, narrowing the addressable set. Even 'hard' public benchmarks are designed to be eroded: FrontierMath was under 2% solved at its November-2024 launch [14], yet by December 2025 a frontier system solved 40.3% of its Tier 1-3 problems with tools (GPT-5.2) — illustrating that a 'currently unsolved' benchmark is a moving target rather than a durable moat; 'genuinely proprietary' is. The contamination evidence sharpens this: on SWE-bench Verified, models identify the buggy file from the issue text alone at up to 76% in-benchmark but only ~53% on repositories outside the set, pointing to memorization rather than reasoning [19] — so even the headline public-set gains overstate transferable capability, while contamination-resistant successors built on private, actively-maintained code (SWE-Bench Pro) hold the best commercial model to a 17.8% resolve rate [18]. Finally, model-collapse severity is contested in magnitude by follow-up work (mixing or accumulating real data avoids collapse [12]), so the 'web tail erosion' mechanism is directional, not a precise rate.

Conclusion

Deduplication, diversity selection, coverage-aware pruning, and model collapse are one phenomenon viewed four ways: all reduce to the coverage of rare regions of the data manifold. The open web is a head-light, tail-heavy distribution whose removable redundancy is large but finite and nearly exhausted, while its tail — the specialized verticals — is where the residual marginal value of data now concentrates [1][2][4][5][11].

The data wall caps fresh head supply and synthetic contamination erodes the web tail [6][9], so the only durable source of new tail coverage is real, human-grounded vertical data. This converts 'underrepresented vertical' from a description of a niche into the precise selection criterion for the highest-marginal-value data to create. The scarce good is rare-tail coverage; the tails are the verticals; and that is where data creation should aim.

INVESTMENT THESIS

The scarce, appreciating input in AI is rare-tail coverage, and the tails are proprietary verticals. Three forces converge: the public-text data wall caps fresh head supply (~300T-token effective stock, 90% CI 100T-1000T, exhausted 2026-2032, median ~2028 [9]); web curation has finite, one-shot upside (SemDeDup already removes 50% of a corpus once [4], and pruning bottoms out at a 24-46% irreducible residual [5]); and synthetic recursion actively erodes the web's tail (model collapse: OPT-125M perplexity rises ~34->50+ within ~5 recursive generations [6]). Meanwhile the capability deficit is demonstrably in the tail — BLOOM-176B QA accuracy climbs from 25% to >55% as a fact's relevant-document count rises from 10^1 to 10^4 , scaling only log-linearly in model size ($R^2=0.98$) [11]. The public-vs-proprietary contrast is the clincher: public-tail verticals are learnable from the web and close fast — SWE-bench Verified went from a 1.96% best system at release [13] to ~80% by late 2025 (Claude Opus 4.5 80.9%, GPT-5.2 80.0%) via RL on open code [15], self-improving agents [16], and open agentic models [17] — and even hard public benchmarks erode (FrontierMath: <2% at its November-2024 launch [14], 40.3% with tools by December 2025). Only genuinely proprietary, contamination-resistant tails stay open: on a successor built from private, actively-maintained repos (SWE-Bench Pro), the best commercial model still resolves only 17.8% [18], and SWE-bench's own scores are partly memorization (76% in-benchmark vs ~53% out-of-benchmark file-path ID) [19]. Bet on companies that own a real, proprietary vertical tail that cannot be scraped, synthesized, or deduplicated away. Underrepresentation on the web is the buy signal, not the risk. Diligence frontier: not whether tail coverage is valuable (the literature settles that) but whether a given vertical's tail data is legally and economically obtainable.

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